

SELECTING REGRESSORS USING THE BAYESIAN INFORMATION CRITERION (BIC)

ABSTRACT. In the context of linear regression and OLS, we discuss information-criteria-based approaches for selecting relevant regressors out of a list of potential regressors. We discuss consistency and oracle properties, and post-selection inference.

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1. SELECTING REGRESSORS

We will discuss the problem of selecting relevant regressors in the context of the linear regression model. However, the procedures discussed here can be generalized and similarly applied to nonlinear models such as logit, GMM, etc. Consider a linear regression model with k *potential regressors*:

$$(1.1) \quad Y_i = \sum_{j=1}^k \beta_j X_{i,j} + U_i,$$

$$E[X_{i,j}U_i] = 0, \quad j = 1, \dots, k.$$

For now, we assume that the number of potential regressors is small: k is fixed and does not depend on n .

Let the set $\mathcal{A}$ denote the list of regressors with nonzero coefficients:

$$\mathcal{A} = \{j : \beta_j \neq 0\}.$$

For example, $\mathcal{A} = \{1, 3, 7\}$ implies that only the regressors $X_{i,1}$, $X_{i,3}$, and $X_{i,7}$ have nonzero coefficients, and that the remaining regressors have coefficients equal to zero. We use $\mathcal{A}_0$ to denote the *true set of relevant regressors*: that is, the true data-generating process (DGP) for Y_i only includes the regressors in $\mathcal{A}_0$:

$$Y_i = \sum_{j \in \mathcal{A}_0} \beta_j X_{i,j} + U_i.$$

Our goal is to estimate $\mathcal{A}_0$ using the data $\{(Y_i, X_i^\top)^\top, i = 1, \dots, n\}$. We use $\hat{\mathcal{A}}_n$ to denote an estimated set of relevant regressors produced by a selection procedure. We say that the selection procedure is consistent if

$$(1.2) \quad \Pr(\hat{\mathcal{A}}_n = \mathcal{A}_0) \rightarrow 1$$

as $n \rightarrow \infty$.

Let $\beta = (\beta_1, \dots, \beta_k)^\top$, and let $\beta_{\mathcal{A}}$ denote the subvector of β that includes only the coefficients in $\mathcal{A}$:

$$\beta_{\mathcal{A}} = (\beta_j : j \in \mathcal{A}).$$

We use $|\mathcal{A}|$ to denote the number of elements in $\mathcal{A}$, and hence $\beta_{\mathcal{A}}$ is a $|\mathcal{A}|$ -subvector of the k -vector β .

Suppose a procedure produced a set of selected regressors $\hat{\mathcal{A}}_n$ and vector of estimates $\hat{\beta}_n = (\hat{\beta}_{n,1}, \dots, \hat{\beta}_{n,k})^\top$. It is natural to set $\hat{\beta}_{n,j} = 0$ for $j \notin \hat{\mathcal{A}}_n$. We say that the procedure is *oracle* if in addition to the consistency property in (1.2),

$$\sqrt{n}(\hat{\beta}_{\mathcal{A}_0} - \beta_{\mathcal{A}_0}) \rightarrow_d N(0, V(\mathcal{A}_0)),$$

where $V(\mathcal{A}_0)$ is the best asymptotic variance one can obtain when the true model $\mathcal{A}_0$ is known. The oracle property means that not only does the econometrician consistently select the true regressors, but also the coefficients on the relevant regressors are estimated as precisely as when the set of true relevant regressors in the DGP is known.

2. BIC

Recall that if the econometrician tries to select the regressors by minimizing the sample sum of squared residuals (SSR), or equivalently maximizing R^2 , the procedure results in overfitting: the SSR is monotonically nonincreasing in the number of included regressors. The idea behind BIC is to penalize the SSR for model complexity.

Let $X_i = (X_{i,1}, \dots, X_{i,k})^\top$, and define $X_{i,\mathcal{A}}$ as the subvector of X_i that includes only the regressors in $\mathcal{A}$:

$$X_{i,\mathcal{A}} = (X_{i,j} : j \in \mathcal{A}).$$

Again, $X_{i,\mathcal{A}}$ is a $|\mathcal{A}|$ -subvector of the k -vector X_i . The true DGP can now be written as

$$\begin{aligned} Y_i &= \sum_{j \in \mathcal{A}_0} \beta_j X_{i,j} + U_i \\ &= X_{i,\mathcal{A}_0}^\top \beta_{\mathcal{A}_0} + U_i. \end{aligned}$$

Let $\hat{\beta}_{n,\mathcal{A}}(\mathcal{A})$ denote the OLS estimator of $\beta_{\mathcal{A}}$ that only uses the regressors in $\mathcal{A}$:

$$\hat{\beta}_{n,\mathcal{A}}(\mathcal{A}) = \left(\sum_{i=1}^n X_{i,\mathcal{A}} X_{i,\mathcal{A}}^\top \right)^{-1} \sum_{i=1}^n X_{i,\mathcal{A}} Y_i.$$

We can set

$$\hat{\beta}_{n,\mathcal{A}^c}(\mathcal{A}) = 0,$$

and view $\hat{\beta}_n(\mathcal{A}) = (\hat{\beta}_{n,\mathcal{A}}(\mathcal{A})^\top, \hat{\beta}_{n,\mathcal{A}^c}(\mathcal{A})^\top)^\top$ as the estimator of $\beta = (\beta_{\mathcal{A}}^\top, \beta_{\mathcal{A}^c}^\top)^\top$ under the model $\mathcal{A}$. The corresponding SSR is given by

$$SSR_n(\mathcal{A}) = \sum_{i=1}^n \left(Y_i - X_{i,\mathcal{A}}^\top \hat{\beta}_{n,\mathcal{A}}(\mathcal{A}) \right)^2.$$

The complexity of the model $\mathcal{A}$ can be measured by the number of included regressors, i.e., the number of elements in $\mathcal{A}$. BIC for the model $\mathcal{A}$ is defined as

$$BIC_n(\mathcal{A}) = SSR_n(\mathcal{A}) + |\mathcal{A}| \log n,$$

where the second term is a *penalty term*. A model with more included regressors receives a larger penalty. A BIC-based selection procedure selects the regressors by minimizing BIC across all possible models:

$$\hat{\mathcal{A}}_n^{BIC} = \arg \min_{\mathcal{A}} BIC_n(\mathcal{A}).$$

We show below that BIC selects the relevant regressors consistently.

Proposition 2.1. *Suppose that data are i.i.d., $E[X_i X_i^\top]$ and $E[U_i^2 X_i X_i^\top]$ are finite and positive definite, and $E[U_i^2] < \infty$. Then $\Pr(\hat{\mathcal{A}}_n^{BIC} = \mathcal{A}_0) \rightarrow 1$ as $n \rightarrow \infty$.*

Proof. It suffices to show that for all $\mathcal{A} \neq \mathcal{A}_0$

$$(2.1) \quad \Pr(BIC_n(\mathcal{A}) > BIC_n(\mathcal{A}_0)) \rightarrow 1,$$

i.e., the true model $\mathcal{A}_0$ minimizes BIC with probability approaching one.

First, consider the *average* SSR for the true model:

$$\begin{aligned} n^{-1} SSR_n(\mathcal{A}_0) &= n^{-1} \sum_{i=1}^n \left(Y_i - X_{i,\mathcal{A}_0}^\top \hat{\beta}_{n,\mathcal{A}_0}(\mathcal{A}_0) \right)^2 \\ &= n^{-1} \sum_{i=1}^n \left(U_i - X_{i,\mathcal{A}_0}^\top (\hat{\beta}_{n,\mathcal{A}_0}(\mathcal{A}_0) - \beta_{\mathcal{A}_0}) \right)^2 \end{aligned}$$

$$\begin{aligned}
&= n^{-1} \sum_{i=1}^n U_i^2 + (\hat{\beta}_{n,\mathcal{A}_0}(\mathcal{A}_0) - \beta_{\mathcal{A}_0})^\top \left(n^{-1} \sum_{i=1}^n X_{i,\mathcal{A}_0} X_{i,\mathcal{A}_0}^\top \right) (\hat{\beta}_{n,\mathcal{A}_0}(\mathcal{A}_0) - \beta_{\mathcal{A}_0}) \\
&\quad - 2 \left(n^{-1} \sum_{i=1}^n X_{i,\mathcal{A}_0} U_i \right)^\top (\hat{\beta}_{n,\mathcal{A}_0}(\mathcal{A}_0) - \beta_{\mathcal{A}_0}) \\
&= \mathbb{E} [U_i^2] + o_p(1),
\end{aligned}$$

where the $o_p(1)$ term in the last line is by the LLN and consistency of the OLS estimator under the true model:

$$\begin{aligned}
n^{-1} \sum_{i=1}^n U_i^2 &= \mathbb{E} [U_i^2] + o_p(1), \\
\hat{\beta}_{n,\mathcal{A}_0} &= \beta_{\mathcal{A}_0} + o_p(1), \\
n^{-1} \sum_{i=1}^n X_{i,\mathcal{A}_0} X_{i,\mathcal{A}_0}^\top &= \mathbb{E} [X_{i,\mathcal{A}_0} X_{i,\mathcal{A}_0}^\top] + o_p(1), \\
n^{-1} \sum_{i=1}^n X_{i,\mathcal{A}_0} U_i &= o_p(1).
\end{aligned}$$

Suppose that a model $\mathcal{A}$ omits some relevant regressors:

$$(\mathcal{A} \cap \mathcal{A}_0) \neq \mathcal{A}_0.$$

Since the OLS estimator is in general inconsistent when there are omitted relevant regressors,

$$\hat{\beta}_n(\mathcal{A}) - \beta \rightarrow_p \delta \neq 0,$$

where $\hat{\beta}_{n,j}(\mathcal{A})$ is the corresponding element of $\hat{\beta}_{n,\mathcal{A}}(\mathcal{A})$ for $j \in \mathcal{A}$, and $\hat{\beta}_n(\mathcal{A}) = 0$ for $j \notin \mathcal{A}$. We have:

$$\begin{aligned}
n^{-1} SSR_n(\mathcal{A}) &= n^{-1} \sum_{i=1}^n \left(Y_i - X_i^\top \hat{\beta}_n(\mathcal{A}) \right)^2 \\
&= n^{-1} \sum_{i=1}^n \left(U_i - X_i^\top (\hat{\beta}_n(\mathcal{A}) - \beta) \right)^2
\end{aligned}$$

$$\begin{aligned}
&= n^{-1} \sum_{i=1}^n U_i^2 + (\hat{\beta}_n(\mathcal{A}) - \beta)^\top \left(n^{-1} \sum_{i=1}^n X_i X_i^\top \right) (\hat{\beta}_n(\mathcal{A}) - \beta) \\
&\quad - 2 \left(n^{-1} \sum_{i=1}^n X_i U_i \right)^\top (\hat{\beta}_n(\mathcal{A}) - \beta) \\
&= \mathbb{E} [U_i^2] + \delta^\top \mathbb{E} [X_i X_i^\top] \delta + o_p(1).
\end{aligned}$$

Note also that

$$|\mathcal{A}| \frac{\log n}{n} = o(1).$$

Therefore, for such a model $\mathcal{A}$,

$$\begin{aligned}
\Pr(BIC_n(\mathcal{A}) > BIC_n(\mathcal{A}_0)) &= \Pr(n^{-1} BIC_n(\mathcal{A}) > n^{-1} BIC_n(\mathcal{A}_0)) \\
&= \Pr\left(n^{-1} SSR_n(\mathcal{A}) + |\mathcal{A}| \frac{\log n}{n} > n^{-1} SSR_n(\mathcal{A}_0) + |\mathcal{A}_0| \frac{\log n}{n}\right) \\
&= \Pr(\delta^\top \mathbb{E} [X_i X_i^\top] \delta + o_p(1) + o(1) > 0) \\
&\rightarrow 1,
\end{aligned}$$

where convergence in the last line holds because $\delta \neq 0$ and $\mathbb{E} [X_i X_i^\top]$ is positive definite.

Next, consider a model $\mathcal{A}$ such that

$$\mathcal{A}_0 \subset \mathcal{A}.$$

In this case, $\mathcal{A}$ contains all the relevant regressors as well as some irrelevant ones. The OLS estimator $\hat{\beta}_{n,\mathcal{A}}(\mathcal{A})$ is consistent and asymptotically normal:

$$n^{1/2}(\hat{\beta}_{n,\mathcal{A}}(\mathcal{A}) - \beta_{\mathcal{A}}) \rightarrow_d \Psi_{\mathcal{A}},$$

where

$$\begin{aligned}
\Psi_{\mathcal{A}} &\sim N(0, V(\mathcal{A})), \\
V(\mathcal{A}) &= (\mathbb{E} [X_{i,\mathcal{A}} X_{i,\mathcal{A}}^\top])^{-1} \mathbb{E} [U_i^2 X_{i,\mathcal{A}} X_{i,\mathcal{A}}^\top] (\mathbb{E} [X_{i,\mathcal{A}} X_{i,\mathcal{A}}^\top])^{-1}.
\end{aligned}$$

The result follows from

$$n^{-1/2} \sum_{i=1}^n X_{i,\mathcal{A}} U_i \rightarrow_d \Phi_{\mathcal{A}},$$

where

$$\Phi_{\mathcal{A}} \sim N(0, E[U_i^2 X_{i,\mathcal{A}} X_{i,\mathcal{A}}^\top]).$$

We have:

$$\begin{aligned} SSR_n(\mathcal{A}) - \sum_{i=1}^n U_i^2 &= \sum_{i=1}^n \left(U_i - X_{i,\mathcal{A}}^\top (\hat{\beta}_{n,\mathcal{A}}(\mathcal{A}) - \beta_{\mathcal{A}}) \right)^2 - \sum_{i=1}^n U_i^2 \\ &= n^{1/2} (\hat{\beta}_{n,\mathcal{A}}(\mathcal{A}) - \beta_{\mathcal{A}})^\top \left(n^{-1} \sum_{i=1}^n X_{i,\mathcal{A}} X_{i,\mathcal{A}}^\top \right) n^{1/2} (\hat{\beta}_{n,\mathcal{A}}(\mathcal{A}) - \beta_{\mathcal{A}}) \\ &\quad - 2 \left(n^{-1/2} \sum_{i=1}^n X_{i,\mathcal{A}} U_i \right)^\top n^{1/2} (\hat{\beta}_{n,\mathcal{A}}(\mathcal{A}) - \beta_{\mathcal{A}}) \\ &\rightarrow_d \Psi_{\mathcal{A}}^\top (E[X_{i,\mathcal{A}} X_{i,\mathcal{A}}^\top]) \Psi_{\mathcal{A}} - 2 \Phi_{\mathcal{A}}^\top \Psi_{\mathcal{A}} \\ &= O_p(1). \end{aligned}$$

By the same arguments,

$$\begin{aligned} SSR_n(\mathcal{A}_0) - \sum_{i=1}^n U_i^2 &\rightarrow_d \Psi_{\mathcal{A}_0}^\top (E[X_{i,\mathcal{A}_0} X_{i,\mathcal{A}_0}^\top]) \Psi_{\mathcal{A}_0} - 2 \Phi_{\mathcal{A}_0}^\top \Psi_{\mathcal{A}_0} \\ &= O_p(1). \end{aligned}$$

Lastly, when $\mathcal{A}_0 \subset \mathcal{A}$,

$$\begin{aligned} \Pr(BIC_n(\mathcal{A}) > BIC_n(\mathcal{A}_0)) &= \Pr(SSR_n(\mathcal{A}) - SSR_n(\mathcal{A}_0) > (|\mathcal{A}_0| - |\mathcal{A}|) \log n) \\ &= \Pr(O_p(1) > (|\mathcal{A}_0| - |\mathcal{A}|) \log n) \\ &\rightarrow 1, \end{aligned}$$

where convergence in the last line holds since $|\mathcal{A}_0| < |\mathcal{A}|$, and therefore

$$(|\mathcal{A}_0| - |\mathcal{A}|) \log n \rightarrow -\infty.$$

□

3. POST-BIC INFERENCE

Suppose the econometrician selects the true model using $\hat{\mathcal{A}}_n^{BIC}$ and conducts inference using $\hat{\beta}_n(\hat{\mathcal{A}}_n^{BIC})$. For $j \in \hat{\mathcal{A}}_n^{BIC}$, the distribution of the estimator of the j th coefficient is given by

$$\begin{aligned}
& \Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\hat{\mathcal{A}}_n^{BIC}) - \beta_j) \leq u \right) \\
&= \Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\hat{\mathcal{A}}_n^{BIC}) - \beta_j) \leq u, \hat{\mathcal{A}}_n^{BIC} = \mathcal{A}_0 \right) \\
&\quad + \Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\hat{\mathcal{A}}_n^{BIC}) - \beta_j) \leq u, \hat{\mathcal{A}}_n^{BIC} \neq \mathcal{A}_0 \right) \\
&= \Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\hat{\mathcal{A}}_n^{BIC}) - \beta_j) \leq u, \hat{\mathcal{A}}_n^{BIC} = \mathcal{A}_0 \right) + o(1) \\
&= \Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\hat{\mathcal{A}}_n^{BIC}) - \beta_j) \leq u \mid \hat{\mathcal{A}}_n^{BIC} = \mathcal{A}_0 \right) \Pr \left(\hat{\mathcal{A}}_n^{BIC} = \mathcal{A}_0 \right) + o(1) \\
&= \Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\mathcal{A}_0) - \beta_j) \leq u \right) (1 + o(1)) + o(1) \\
&= \Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\mathcal{A}_0) - \beta_j) \leq u \right) + o(1).
\end{aligned}$$

The second equality holds by

$$\Pr \left(n^{1/2}(\hat{\beta}_{n,j}(\hat{\mathcal{A}}_n^{BIC}) - \beta_j) \leq u, \hat{\mathcal{A}}_n^{BIC} \neq \mathcal{A}_0 \right) \leq \Pr \left(\hat{\mathcal{A}}_n^{BIC} \neq \mathcal{A}_0 \right) = o(1).$$

Hence, the BIC-based selection and estimation procedure is an oracle procedure.

4. AKAIKE INFORMATION CRITERION (AIC)

AIC is another popular criterion for model selection (and actually precedes BIC). AIC for a model $\mathcal{A}$ is defined as

$$AIC_n(\mathcal{A}) = SSR_n(\mathcal{A}) + 2|\mathcal{A}|.$$

Compared with BIC, AIC penalizes model complexity less heavily and therefore tends to select a larger model with more regressors.

By the same arguments as in the proof of Proposition 2.1, for a model $\mathcal{A}$ that omits some relevant regressors, i.e., $(\mathcal{A} \cap \mathcal{A}_0) \neq \mathcal{A}_0$,

$$\Pr(AIC_n(\mathcal{A}) > AIC_n(\mathcal{A}_0)) \rightarrow 1.$$

However, because the AIC penalty is not sufficiently strong, if $\mathcal{A}_0 \subset \mathcal{A}$,

$$\Pr(AIC_n(\mathcal{A}) > AIC_n(\mathcal{A}_0)) \nrightarrow 1.$$

Hence, while AIC detects omitted regressors with probability approaching one, it is more likely than BIC to overfit by including some irrelevant regressors.

5. LIMITATIONS

Our arguments have several limitations. First, we assumed that k is small (fixed) and some of our arguments do not apply when the number of potential regressors is comparable to the sample size. However, this technical issue can be resolved with somewhat different arguments.

More importantly, our analysis ignores the situation in which some β_j , while nonzero, are very close to zero. It is unreasonable to expect that BIC (or any other procedure) can detect such small coefficients with a probability approaching one. Thus, even in the limit, regressors with very small coefficients are likely to be omitted from the model, which can potentially create an omitted variable bias. This shortcoming can be addressed using a double selection procedure, which will be discussed later in the context of Lasso.

Lastly, the BIC procedure may be infeasible in practice if the number of potential regressors is very large. There are 2^k possible models $\mathcal{A}$, and if $k = 30$, one has to run and compare over 1 billion potential regressions. For $k = 40$, one has to run over 1 trillion models. For example, suppose that the econometrician considers flexible specifications that include quadratic terms as well as pairwise interaction terms of the

right-hand side variables. In that case, 10 potential right-hand side variables generate 65 potential regressors.